



Predicts a student's **employability score** (0–100) and recommends one of **six career paths** from a 33-field academic / technical / soft-skill / activity profile, using an ensemble ML pipeline (scikit-learn + XGBoost) served through FastAPI, with real per-prediction **SHAP explainability**, confidence across all six careers, and an LLM chat assistant for follow-up guidance — presented through a React dashboard.

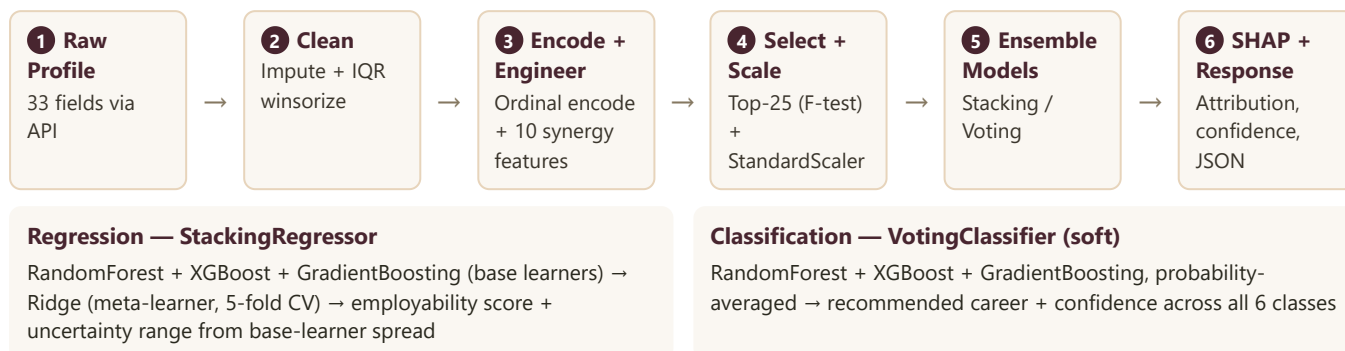
TECH STACK

Backend	Python, FastAPI, Uvicorn, Pydantic
ML / Data	scikit-learn, XGBoost, SHAP, pandas, numpy, pyarrow
EDA	matplotlib, seaborn
Chat / LLM	Hugging Face Inference (huggingface_hub), Qwen2.5-7B-Instruct
Frontend	React 19, Vite, Tailwind CSS v4
Animation / Charts	Framer Motion, hand-rolled inline SVG (no chart library)
Explainability	SHAP TreeExplainer, z-score population comparison

DATASET & TARGETS

Rows	303 (post-dedup)
Raw features	33 (academic, technical, soft-skill, activity)
Engineered features	+10 derived (synergy / index scores)
Selected per model	Top 25 of 41 (SelectKBest)
Regression target	Employability_Score (0–100)
Classification target	Recommended_Career_Path (6 classes)

PIPELINE WORKFLOW



REGRESSION ACCURACY — EMPLOYABILITY SCORE

MODEL	R ²	MAE	RMSE
Ridge Regression	0.9315	2.490	2.896
Random Forest	0.9401	2.160	2.708
XGBoost	0.9484	2.025	2.514
Stacking Ensemble (deployed)	0.9442	2.142	2.613

CLASSIFICATION ACCURACY — CAREER PATH

MODEL	ACCURACY	F1 (MACRO)	PRECISION
Voting Ensemble (deployed)	95.08%	0.9519	0.9577

6 classes: Cybersecurity Analyst · Data Analyst · Machine Learning Engineer · Software Engineer · UI/UX Designer · Web Developer

KEY FEATURES

- Real confidence, not just a label** — soft-voting exposes probability across all 6 careers, plus an uncertainty range on the score from base-learner disagreement.
- Genuine explainability** — SHAP TreeExplainer attribution (not a canned explanation) alongside population z-score comparison, toggleable in the UI, plus a SHAP waterfall chart.
- Bias-audited** — Gender was originally an input feature; audited, found to contribute zero predictive value, and removed — retrained metrics identical to 4 decimal places.
- LLM chat assistant** — free-tier Hugging Face model answers follow-up questions grounded in the actual prediction.